

Agentic Configuration of Model Predictive Control for Launch Vehicle Failure Mitigation

16.S897 Aerospace Systems: Class Project Proposal
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Motivation & Objective

Launch vehicle propulsion failures can cause sudden changes in thrust magnitude, thrust symmetry, and available control authority during ascent. The appropriate vehicle response depends on the nature and severity of the failure: a mild thrust degradation may permit continued flight with trajectory adjustment, whereas a severe or asymmetric thrust loss may require retargeting, tighter control constraints, or prioritization of vehicle stability. Model Predictive Control (MPC) provides a natural framework for this problem because it can optimize control actions while accounting for vehicle dynamics and operational constraints.

However, MPC performance depends strongly on how the optimization problem is configured, including the prediction horizon, objective weights, terminal target, and active constraints. These parameters are typically selected offline, while the appropriate formulation following a propulsion failure may depend on the evolving vehicle and failure state.

This project investigates whether an **agentic AI supervisor can dynamically configure an MPC problem in response to a launch vehicle propulsion failure**. Rather than replacing the controller or directly generating control commands, the agent will operate at the supervisory level: interpreting a structured failure state, selecting from a bounded set of MPC configurations, invoking the MPC solver, evaluating its output, and revising the configuration when necessary. The broader objective is to evaluate whether agentic reasoning can provide an adaptive interface between changing system conditions and deterministic model-based control.

Research Question

Can an agentic AI supervisor use uncertain propulsion-health information and model feedback to configure an appropriate MPC problem for launch vehicle failure mitigation while preserving hard vehicle constraints?

Proposed Framework

The proposed architecture is:

Failure-State Estimate $\rightarrow$ Agentic Supervisor $\rightarrow$ MPC Specification $\rightarrow$ MPC Solver $\rightarrow$ Verifier

The verifier returns solver and constraint feedback to the agent, creating an iterative *observe–configure–solve–evaluate–revise* loop.

The input will consist of synthetic structured states such as failure type, onset time, residual thrust or specific impulse, severity uncertainty, and current vehicle state. The agent may configure:

- prediction and control horizons;
- terminal trajectory or mission target;
- objective emphasis, e.g., trajectory tracking, propellant use, control effort, or attitude stabilization;
- approved constraint templates or constraint tightening; and
- mitigation mode.

The deterministic MPC solver computes the trajectory/control solution. A separate verifier evaluates convergence, feasibility, actuator limits, trajectory constraints, and terminal performance. Failed checks are returned to the agent for bounded replanning.

Methodology & Evaluation

A reduced-order launch vehicle model will be used initially, avoiding dependence on an engine dataset. Test cases will span mild-to-severe thrust degradation, asymmetric or single-engine thrust loss, specific-impulse degradation, different failure-onset times, and uncertain residual propulsion capability. The same plant model and MPC solver will be used for all approaches.

The agentic approach will be compared against three baselines:

- a single fixed MPC formulation;
- a rule-based failure-to-MPC configuration scheduler; and
- a single-shot LLM configuration without solver feedback.

Performance will be evaluated using feasible-solution rate, hard-constraint violations, terminal trajectory error, control effort and actuator saturation, computational cost, number of replanning iterations, and performance relative to the best MPC configuration identified through an offline search over the same approved configuration space.

Expected Contribution

The expected contribution is a **tool-constrained agentic framework for adaptive MPC configuration under changing system conditions**. The project will examine where agentic reasoning can augment conventional optimization by making supervisory decisions about how an MPC problem should be formulated.